



ROOT



MATLAB Geek

Summer Bootcamp | 2026

WiseDuck



Code is poetry written for
machines!

Presented by



Instructor: **mmd**
@mmd1023

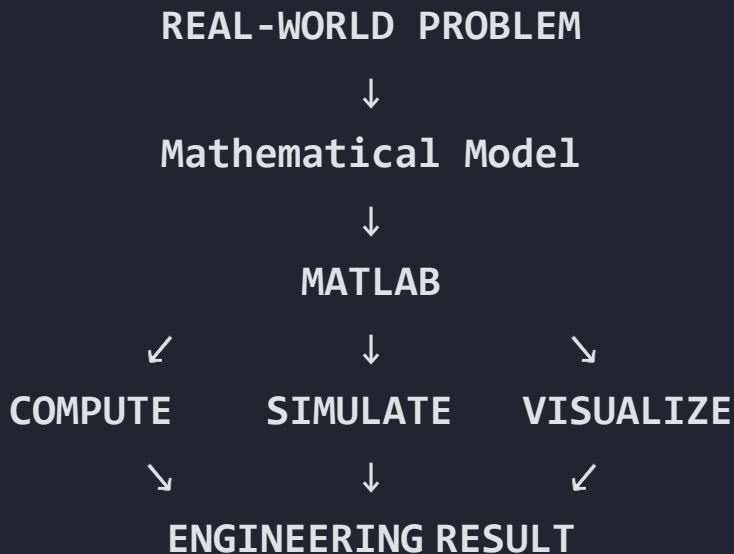


Organizer: **ElectroGeek**
@ElectroGeek255

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Why MATLAB?



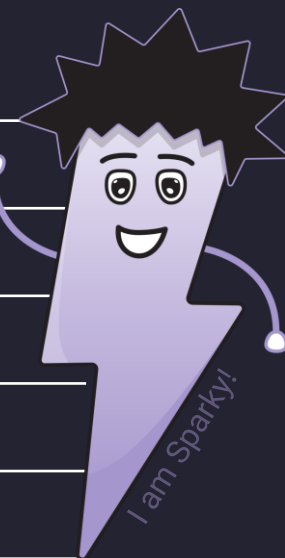
MATLAB connects mathematics, programming, and engineering.

- Engineering Computing
- Matrix & Vector Based
- Visualization
- Simulation & Modeling
- Engineering Ecosystem



MATLAB vs CPP and Python

ویژگی	MATLAB	Python	C/C++
سهولت کدنویسی 	★★★★★	★★★★	★★
سرعت اجرا 	★★★★	★★★	★★★★★
نمودار 	★★★★★	★★★★	★
ریاضیات مهندسی 	★★★★★	★★★★	★★
جعبه ابزارها 	★★★★★	★★★	★
هزینه 	تجاری	رایگان	رایگان



Let's Code!



The Wanderer



Variable

Type	Description	Range
double	Default MATLAB type; double-precision floating	$\pm 1.8 \times 10^{308}$
single	Single-precision floating-point	$\pm 3.4 \times 10^{38}$
int8	Signed 8-bit integer	-128 to 127
uint8	Unsigned 8-bit integer	0 to 255
int16	Signed 16-bit integer	-32,768 to 32,767
uint16	Unsigned 16-bit integer	0 to 65,535
int32	Signed 32-bit integer	-2^{31} to $2^{31} - 1$
uint32	Unsigned 32-bit integer	0 to $2^{32} - 1$
int64	Signed 64-bit integer	-2^{63} to $2^{63} - 1$
uint64	Unsigned 64-bit integer	0 to $2^{64} - 1$
logical	Logical (boolean) values	0 / 1 (false / true)
char	Character	Letters and symbols
string	Text	String of characters



“A variable is just a name for a value. Choose the name wisely, your future self has to debug it.”

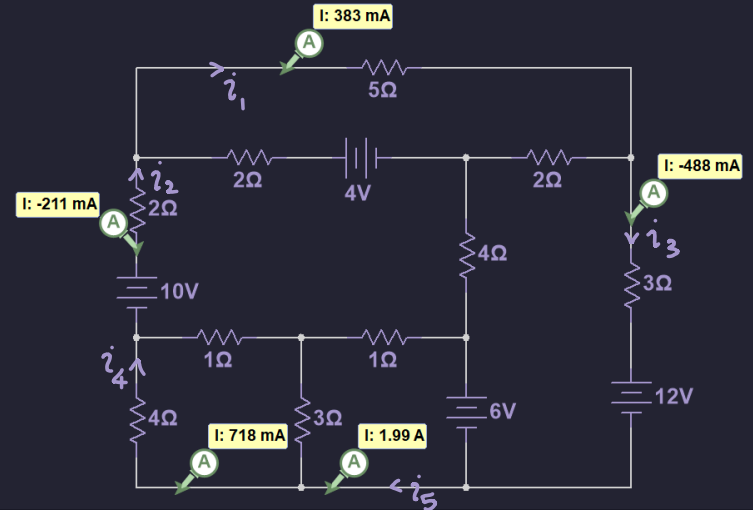
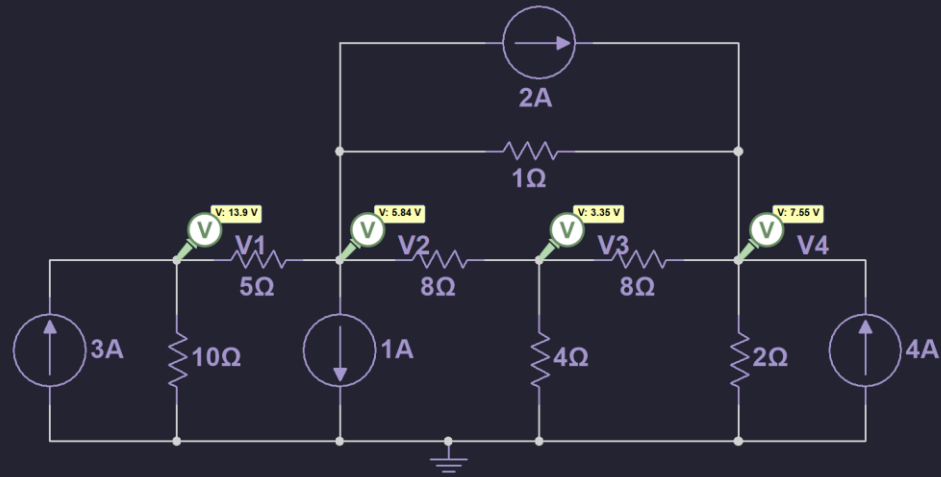


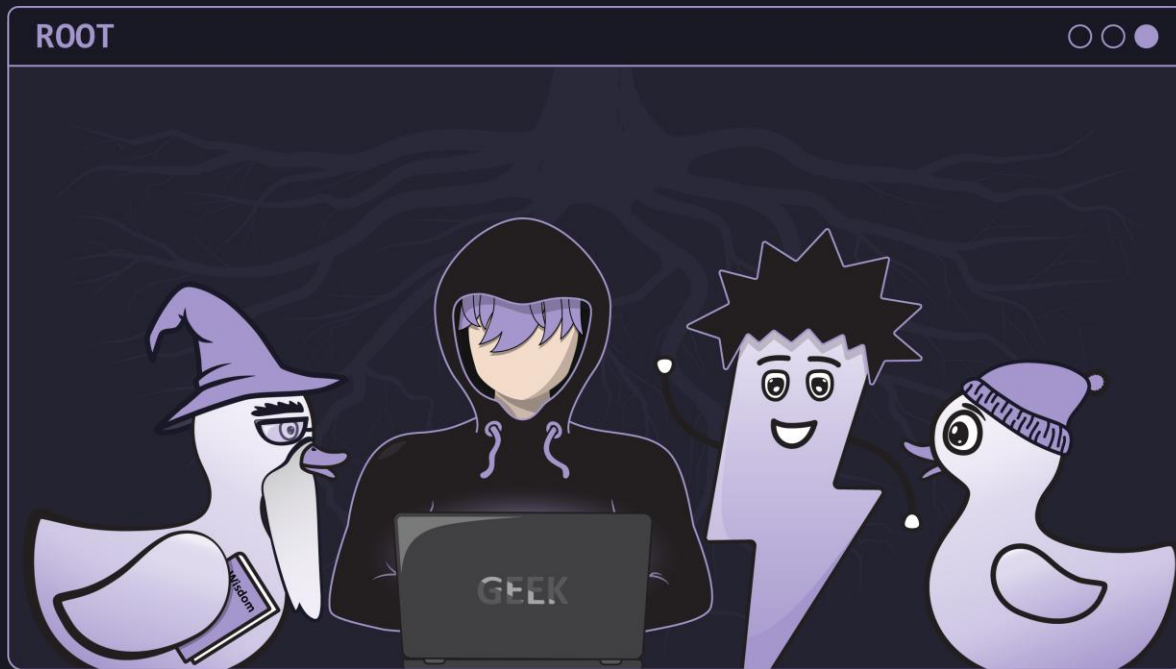
EXAMPLE 0,0 | Linear System

$$\begin{bmatrix} \frac{1}{10} + \frac{1}{5} & -\frac{1}{5} & 0 & 0 \\ -\frac{1}{5} & \frac{1}{5} + \frac{1}{8} + 1 & -\frac{1}{8} & -1 \\ 0 & -\frac{1}{8} & \frac{1}{8} + \frac{1}{8} + \frac{1}{4} & -\frac{1}{8} \\ 0 & -1 & -\frac{1}{8} & \frac{1}{8} + \frac{1}{2} + 1 \end{bmatrix} \times \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 0 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 5 + 2 + 2 & -2 & -2 & 0 & 0 \\ -2 & 2 + 2 + 1 + 1 + 4 & -4 & -1 & -1 \\ -2 & -4 & 2 + 3 + 4 & 0 & 0 \\ 0 & -1 & 0 & 1 + 3 + 4 & -3 \\ 0 & -1 & -1 & 0 & 1 + 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \times \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix}$$







The End!



THANKS!

Do you have any questions?



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<https://github.com/mmd1023>



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